

WeatherForge Documentation

Deploying the Dashboard for a Different State

The Concept

The WeatherForge dashboard is designed to be fully state-agnostic. All Minnesota-specific logic has been removed from the codebase and replaced with a central **STATE_CONFIG** block and a set of swappable data files. Redeploying for any US state requires no code changes beyond updating those three config values and dropping in the appropriate data files.

How It Works

At startup the app reads **STATE_CONFIG["state_fips_prefix"]** and uses it to filter every data source to the correct state automatically:

- The county GeoJSON is filtered to only features whose ID starts with the state prefix.
- The county population CSV loads only rows matching the prefix — a single national file works for all states.
- The map bounding box is derived automatically from the loaded GeoJSON coordinates — no hardcoded lat/lon ranges.
- All UI strings, live alerts API calls, and summary text use the state name and abbreviation from **STATE_CONFIG**.

Why This Approach

This setup is intentionally minimal. Rather than building a runtime state selector into the UI (Option B) or a full national map (Option C), the current architecture treats each state as a separate deployment instance. This keeps the app fast, simple, and easy to demonstrate — and the groundwork is already in place to upgrade to a national multi-state deployment later without a rewrite.

How to Switch States

There are only two steps required to redeploy the dashboard for any US state.

1. Update **STATE_CONFIG** in **app.py**

Locate the **STATE_CONFIG** block near the top of **app.py** and update the three values:

```
STATE_CONFIG: dict[str, str] = { "state_name": "Minnesota", # e.g. "Texas"
    "state_abbr": "MN", # e.g. "TX" "state_fips_prefix": "27", # e.g. "48" }
```

The `state_fips_prefix` is the two-digit US Census FIPS code assigned to each state. All county FIPS codes in the country begin with their state's two-digit prefix — this single value controls which counties are loaded from every data source.

2. Swap the three data files

Replace the following files in the project directory with the equivalent files for the new state. File names do not need to change — the app reads them by their fixed names regardless of which state's data they contain:

- **mn_counties.geojson** — county boundary geometries for the new state
- **storm_events_mn.parquet** — NOAA storm event records for the new state
- **storm_weather_summary_mn.parquet** — storm events joined with daily weather observations

county_populations.csv does NOT need to be swapped. It is designed to hold population data for all states simultaneously. Simply add the new state's county rows to the existing file — the app filters by state prefix automatically.

county_populations.csv Structure

The population file uses a wide format with one row per county and one column per census decade. The fips column is a zero-padded 5-digit string that uniquely identifies every county in the country:

```
fips, 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020 27001, 12162, 12162, 11403,
13404, 12425, 15301, 16202, 15696 27003, 35579, 85916, ... 48001, ..., ..., ...
(Texas rows added here)
```

Source: U.S. Census Bureau decennial census. Values between decades are linearly interpolated by `get_county_population()`. Values outside the 1950-2020 range are extrapolated using the nearest decadal slope. Counties not present in the file fall back to a neutral denominator of 10,000.

What Happens Automatically After Switching

Once `STATE_CONFIG` is updated and the data files are swapped, the following all update without any additional code changes:

- Map viewport — bounds derived from the new GeoJSON, no coordinates to look up
- County filter sidebar — repopulates from the new dataset's county names
- Live Alerts feed — NWS API called with the new state abbreviation
- All UI text — state name appears in titles, summaries, and status messages
- Risk score normalization — population denominator loaded for the new state's counties
- Hazard type dropdowns — rebuilt from whatever event types exist in the new data

See [WeatherForge_LiveFeed_FutureChange.pdf](#) for instructions on switching the Live Feed from Dataset-Anchored (demo) mode to Real-Time (production) mode.